



Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science)

Semester – VI

CDS-364

Lab on CDS-361

Digital Forensic II Laboratory

Savitribai Phule Pune University
T.Y.B.Sc (Cyber and Digital Science) Sem-VI
(CDS 364)
Practical based on
CDS 361 Digital Forensic 2

Time : 3 Hours

Max. Marks: 35

Q.1 Solve the following questions

[15M]

- a) Extract metadata from an email header and analyze its origin and routing.
- b) Identify forged email headers and detect spoofing attempts.
- c) Investigate an email phishing attack using header analysis.

Q.2 Describe the steps you followed to recover contacts using your Google account

[15M]

Q3. Viva

[5M]

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Q 1. Extracting device information of your Android device using ADB. Create a report of your findings.

[15M]

Q2. Use WinHex to find the last access time of Google Drive on your computer?

[15M]

Q3. Viva

[5M]

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Q1. Solve the following questions

[15M]

- a) What insights were gained by using Wireshark to generate a list of the top 10 websites visited by users in the network?

Q2. Solve the following questions

[15M]

- a) Extract a list of all image, video, and audio files from a mobile device.
- b) How can you retrieve metadata (e.g., date created, location) for these files?

Q3. Viva

[5M]

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Q1. Open an email and extract its header. Identify the following: [15M]

- a) Sender's email address
- b) Recipient's email address
- c) Date and time of sending
- d) Email servers involved in transmission

Q2. Solve the following questions [15M]

- a) How did you use Autopsy to analyze the Android image, and what options does it provide for file recovery?

Q3. Viva [5M]

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Q.1 Solve the following questions [15M]

- a) Use an email tracer tool (e.g., MXToolbox, EmailTrackerPro) to track the sender's IP address.
- b) Perform a forensic examination of a compromised email account.
- c) Analyze Base64 encoded email content and extract hidden data.

Q2. Solve the following questions [15M]

- a) What steps did you follow to extract data from the chosen social media application?

Q3. Viva [5M]

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Q1. Solve the following

[15M]

- a) How did you use the tcpdump command-line program to examine network traffic?

Q2 . Solve the following

[15M]

- a) Document the extracted data, including time of extraction and method used.
- b) What key details should be included in a forensic data report?

Q3. Viva

[5M]

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Q1. Solve the following questions

[15M]

- a) What information can be obtained by examining files associated with VMs, especially log files, in the context of investigating a crime or incident?

Q2. Solve the following

[15M]

- a) Identify the SPF (Sender Policy Framework) record in an email header. What does it tell you about the legitimacy of the sender?
- b) Find the DKIM (Domain Keys Identified Mail) **signature** in the email header. What domain signed the email, and was it verified successfully?

Q3. Viva

[5M]

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Q.1 Solve the following questions

[15M]

- a) Extracting device information of your Android device using ADB. Create a report of your findings.

Q2. Which tool or method did you use to create an image of your phone data, and why?

[15M]

Q3. Viva

[5M]

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Q.1 Solve the following questions

[15M]

- a) Using Wireshark, how do you generate a list of the top 10 websites accessed from a VM?
- b) What indicators can be used to detect data exfiltration, unauthorized access, or malware communication within a network?

Q2. Solve the following questions

[15M]

- a) How can you extract and analyze call logs from the mobile device image?
- b) How to recover deleted data can be recovered from the mobile image

Q3. Viva

[5M]

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Q.1 Solve the following questions

[15M]

- a) How can you use ADB (Android Debug Bridge) to retrieve basic device information such as manufacturer, model, and OS version?
- b) What ADB command retrieves a list of installed applications on an Android device?

Q2. Solve the following questions

[15M]

- a) Look for the DMARC (Domain-based Message Authentication, Reporting & Conformance) record. What policy is set for handling failed authentication?
- b) In what time zone does the sender's email server reside? What header field provides this information?

Q3. Viva

[5M]

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Q.1 Solve the following questions

[15M]

- a) How can you use the tcpdump command-line tool to capture and analyze network packets?
- b) How can you filter network packets to find evidence of suspicious activity within a VM?

Q2. Solve the following questions

[15M]

- a) Investigate email logs on a mail server using forensic tools.
- b) Use Wireshark to capture and analyze SMTP traffic for forensic evidence.
- c) Extract and decode MIME-encoded email messages.

Q3. Viva

[5M]

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Q.1 Solve the following questions

[15M]

- a) How can you extract and analyze metadata from images and videos found in the mobile device image

Q2. Solve the following questions

[15M]

- a) What permissions are required to access system-level information using ADB?
- b) What security risks are associated with enabling ADB on a mobile device?

Q3. Viva

[5M]

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Q1. Solve the following questions

[15M]

- a) Analyze social media data for potential cyber threats (e.g., phishing, fake news).
- b) What techniques can be used to identify malicious content?
- c) Extract and analyze a suspect's social media activity

Q2. Solve the following questions

[15M]

- a) How can Google backups be used to restore deleted contacts, and what limitations exist?
- b) What security and privacy concerns should be considered when recovering deleted mobile data?

Q3. Viva

[5M]

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Q.1 Solve the following questions

[15M]

- a) Write a script to extract public profile details (name, bio, location) from a social media platform (e.g., Twitter, Instagram, Facebook).
- b) What API endpoints can be used for this purpose?

Q2.Solve the following questions

[15M]

- a) Capture live network traffic using Wireshark and analyze packets for suspicious activity.
- b) Identify login credentials, malicious requests, or unauthorized data transfers.

Q3. Viva

[5M]

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Q1. Solve the following questions

[15M]

- a) How can you use intrusion detection systems (IDS) like Snort to monitor traffic from a virtual machine?
- b) How can encrypted network traffic be analyzed for forensic evidence in a virtualized environment?

Q2. Solve the following

[15M]

- a) How do you create an Android device image for forensic analysis?
- b) What steps are involved in loading and analyzing a mobile device image

Q3. Viva

[5M]

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Q1. Solve the following questions

[15M]

- a) Analyze email headers to trace the origin of a phishing or spam email.
- b) Identify spoofed email senders and embedded malicious links.

Q2. Solve the following questions

[15M]

- a) Investigate logs from a hypervisor (e.g., VMware, VirtualBox, Hyper-V) to detect unauthorized access or unusual activities.
- b) Identify timestamps of VM starts, stops, and snapshots.

Q3. Viva

[5M]

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Q1. Solve the following questions

[15M]

- a) Using (MXToolbox, Email Header Analyzer) analyze email header.
- b) Find out SPF, DKIM, and DMARC records, and how do they help in email authentication?

Q2. Solve the following questions

[15M]

- a) How can you acquire a forensic image of a host machine running multiple VMs?
- b) What types of network logs (e.g., firewall, IDS/IPS, system logs) can be used to trace activities between the host and VMs?

Q3. Viva

[5M]

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Q1. Solve the following questions

[15M]

- a) How can you extract VM log files, configuration files, and snapshots for forensic analysis?
- b) What forensic artifacts can be found inside a VM that are different from those on a physical machine?

Q2. Solve the following questions

[15M]

- a) How can you determine the number of times Google Drive has been accessed on a system?
- b) What artifacts are created on a computer when Google Drive is used?

Q3. Viva

[5M]

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Q1. Solve the following questions **[15M]**

- a) What types of network logs (e.g., firewall, IDS/IPS, system logs) can be used to trace activities between the host and VMs?
- b) How can you use system logs to determine when a VM was created, modified, or deleted?

Q2. Solve the following questions **[15M]**

- a) What tools can be used to recover deleted SMS messages from an Android device?
- b) How can you analyze the recovered SMS messages to verify their integrity?

Q3. Viva **[5M]**

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Q1. Solve the following questions

[15M]

- a) How can you identify the originating IP address of an email, and what tool can be used to perform a reverse lookup?

Q2 . Solve the following questions

[15M]

- a) How can you identify the top 10 websites visited by a VM user using Wireshark?
b) What forensic indicators can be found in network packets that reveal communication between a VM and external servers?

Q3. Viva

[5M]

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Q.1 Solve the following questions

[15M]

- a) Extract hidden files and encrypted data stored in a mobile device.
- b) What encryption techniques are commonly used to protect sensitive mobile data?

Q2. Solve the following

[15M]

- a) What types of metadata can be extracted from JPG, PNG, and BMP images using Exif Reader?
- b) How do date, time, and geolocation metadata help in forensic analysis of images?

Q3. Viva

[5M]

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Q1. Solve the following questions

[15M]

- a) How can you extract and analyze VM-related log files to reconstruct user activities and detect suspicious actions?
- b) What network logs can be used to trace interactions between the host machine and a VM?

Q2. Solve the following questions

[15M]

- a) How can you extract and analyze a user's cloud transaction history using WinHex?
- b) What forensic techniques can be used to identify create, modify, and delete timestamps of cloud-stored files?

Q3. Viva

[5M]

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Q1. Solve the following questions

[15M]

- a) Extract email and SMS data from a mobile device to identify phishing messages.
- b) How can metadata from messages help in identifying the attacker?

Q2 . Solve the following questions

[15M]

- a) How can you differentiate between locally modified files and files modified directly in the cloud?
- b) What types of logs and metadata can help in reconstructing a user's cloud activity?

Q3. Viva

[5M]

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Q1. Solve the following questions

[15 M]

- a) How do you acquire a forensic image of a host machine running a VM using FTK Imager Lite?
- b) What types of data (e.g., system logs, network logs, VM disk images) should be collected for forensic analysis?

Q2. Solve the following questions

[15M]

- a) How can WinHex be used to analyze the Windows registry for Google Drive activity?
- b) What specific registry keys store information about the last access time of Google Drive?

Q3. Viva

[5M]

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Q1. Solve the following questions

[15 M]

- a) Extract and examine VM snapshots for forensic evidence.
- b) Recover deleted files from an old snapshot.

Q2. Solve the following questions

[15M]

- a) Analyze email headers to trace the origin of a phishing or spam email.
- b) Identify spoofed email senders and embedded malicious links.

Q3. Viva

[5M]